Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Parallel lines and transversals

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

2 [Foundational] • Target: Parallel lines and transversals

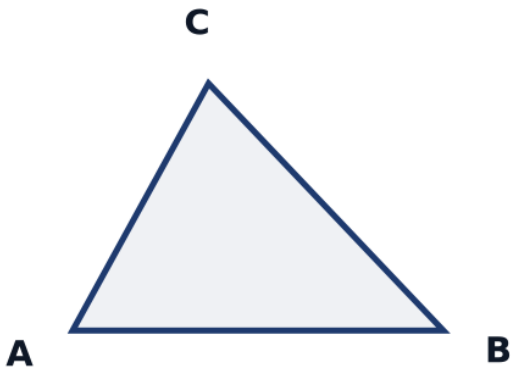
On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

3 [Foundational] • Target: Parallel lines and transversals

In ΔABC studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

4 [Foundational] • Target: Parallel lines and transversals

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.



5 [Foundational] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

6 [Foundational] • Target: Parallel lines and transversals

In ΔABC studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

7 [Foundational] • Target: Parallel lines and transversals

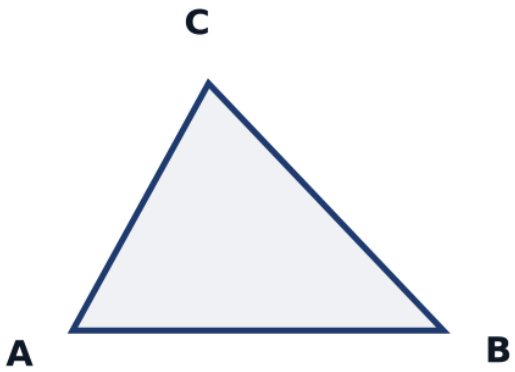
A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

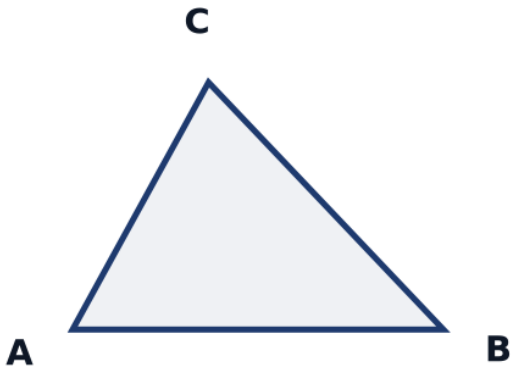
8 **[Intermediate]** • **Target: Parallel lines and transversals**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



9 **[Intermediate]** • **Target: Parallel lines and transversals**

In ΔABC studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



10 **[Intermediate]** • **Target: Parallel lines and transversals**

A geometric figure scaled by factor $k = 5$ has an original side length of $7\ \text{cm}$. What is the corresponding image length? Show your complete algebraic work.

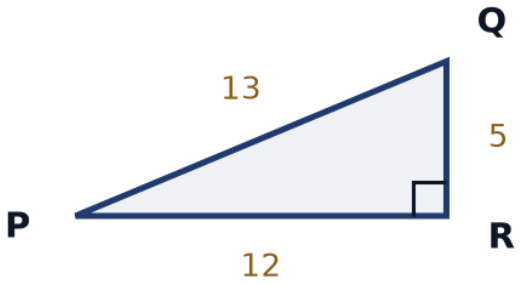
11 **[Intermediate]** • **Target: Parallel lines and transversals**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

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12 [Intermediate] • Target: Triangles

In ΔABC studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



13 [Intermediate] • Target: Triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

14 [Intermediate] • Target: Triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

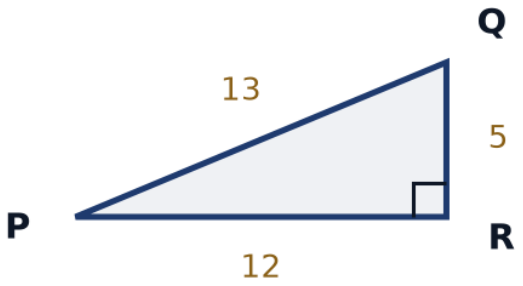
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Triangles • Cross-Lesson: 5.01 Geometry

In ΔABC studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

16 [Challenging] • Target: Triangles • Cross-Lesson: 6.01 Trigonometry

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.



17 [Challenging] • Target: Triangles • Cross-Lesson: 7.01 Transformations & Vectors

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

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18 [Challenging] • Target: Triangles • Cross-Lesson: 8.01 Probability & Statistics

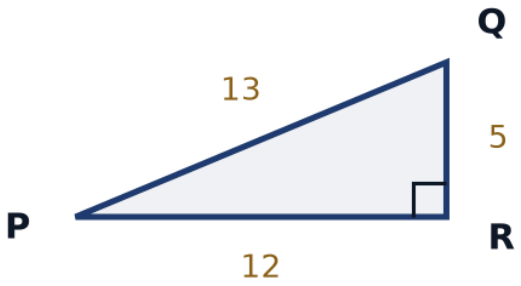
In ΔABC studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

19 [Challenging] • Target: Triangles • Cross-Lesson: 1.01 Number

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

20 [Challenging] • Target: Triangles • Cross-Lesson: 10.01 Sets & Venn Diagrams

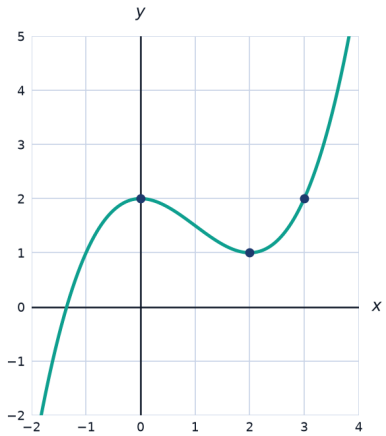
On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Geometrical diagrams

In ΔABC studying geometrical diagrams, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 84°

b. 94°

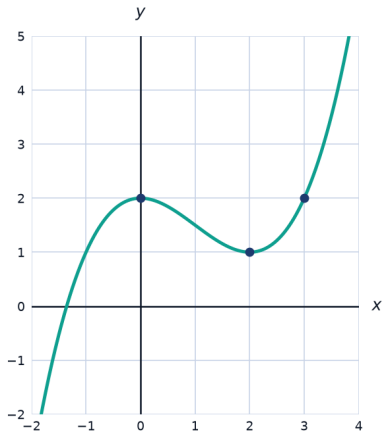
c. 76°

d. 129°

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22 [Foundational] • Target: Geometrical diagrams

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?



a. 12 cm

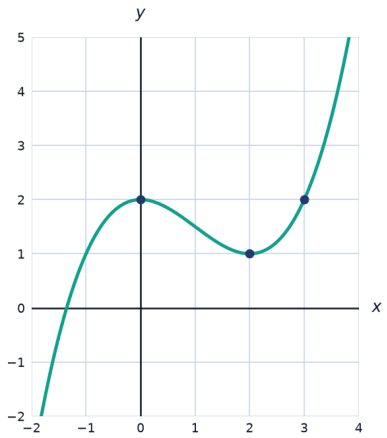
b. 15 cm

c. 20 cm

d. 45 cm

23 [Foundational] • Target: Geometrical diagrams

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?



a. 14

b. 21

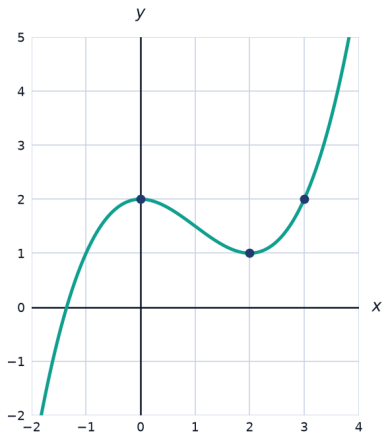
c. 19

d. 24

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24 [Foundational] • Target: Geometrical diagrams

In ΔABC studying geometrical diagrams, $m\angle A = 51^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 63°

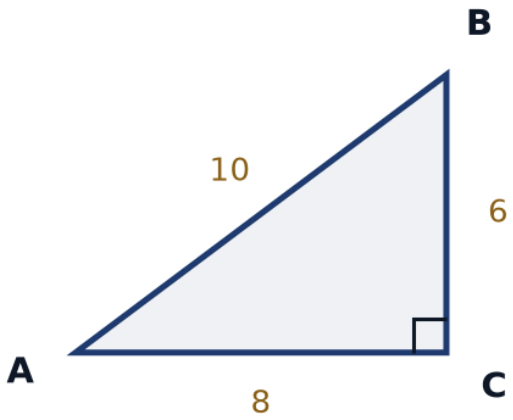
b. 73°

c. 55°

d. 129°

25 [Foundational] • Target: Measuring angles

In ΔABC studying measuring angles, $m\angle A = 52^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 83°

b. 93°

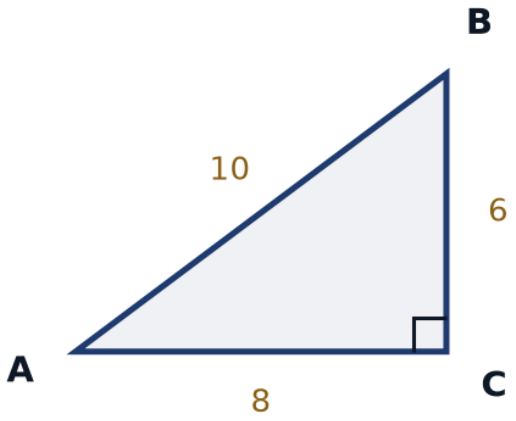
c. 75°

d. 128°

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26 [Foundational] • Target: Measuring angles

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?



a. 15 cm

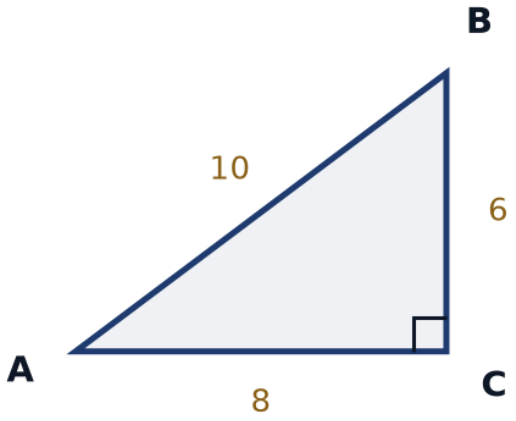
b. 18 cm

c. 24 cm

d. 54 cm

27 [Foundational] • Target: Measuring angles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$?



a. 9

b. 13

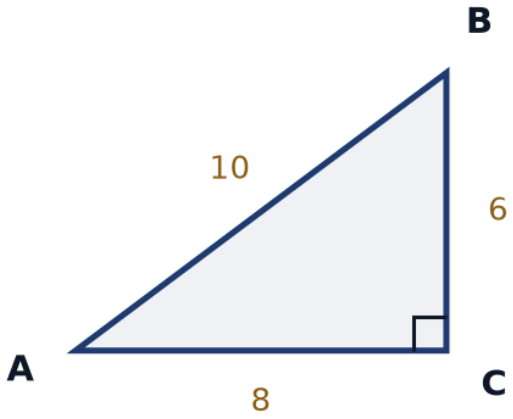
c. 11

d. 13

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28 [Foundational] • Target: Measuring angles

In ΔABC studying measuring angles, $m\angle A = 52^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 62°

b. 72°

c. 54°

d. 128°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Geometrical diagrams

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

30 [Intermediate] • Target: Geometrical diagrams

In ΔABC studying geometrical diagrams, $m\angle A = 51^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 77°

b. 87°

c. 69°

d. 129°

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31 [Intermediate] • Target: Geometrical diagrams

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm

c. 25 cm

d. 80 cm

32 [Intermediate] • Target: Geometrical diagrams

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 19

b. 26

c. 24

d. 29

33 [Intermediate] • Target: Measuring angles

In ΔABC studying measuring angles, $m\angle A = 52^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 76°

b. 86°

c. 68°

d. 128°

34 [Intermediate] • Target: Measuring angles

In ΔABC studying measuring angles, $m\angle A = 52^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 69°

b. 79°

c. 61°

d. 128°

35 [Intermediate] • Target: Measuring angles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 11

b. 15

c. 13

d. 15

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36 [Intermediate] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$?

a. 6

b. 11

c. 9

d. 12

37 [Intermediate] • Target: Parallel lines and transversals

In ΔABC studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 68°

b. 78°

c. 60°

d. 127°

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Parallel lines and transversals • Cross-Lesson: 8.01 Probability & Statistics

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

39 [Challenging] • Target: Triangles • Cross-Lesson: 1.01 Number

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 13

c. 11

d. 15

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40 [Challenging] • Target: Triangles • Cross-Lesson: 10.01 Sets & Venn Diagrams

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 11

b. 17

c. 15

d. 19

41 [Challenging] • Target: Triangles • Cross-Lesson: 2.01 Algebra & Graphs

A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm

c. 48 cm

d. 200 cm

42 [Challenging] • Target: Quadrilaterals • Cross-Lesson: 3.01 Coordinate Geometry

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

43 [Challenging] • Target: Quadrilaterals • Cross-Lesson: 4.01 Mensuration

For the model $f(x) = 2x + 8$ in quadrilaterals, evaluate $f(4)$.

a. 16

b. 18

c. 8

d. 20

44 [Challenging] • Target: Quadrilaterals • Cross-Lesson: 5.01 Geometry

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

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45 [Challenging] • Target: Angles of polygons • Cross-Lesson: 6.01 Trigonometry

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 11

b. 15

c. 13

d. 15

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